



Final Presentation

Development and Validation of a Dexterous Hand System

··· Dexhand Designed by The Robot Studio···

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Overall Workflow

Mechanical Assembly

Electronics and Embedded Control
SystemTkinter Monitor Interface
/Simulation**Project Background**

- DexHand is used in this project as an open-source dexterous hand platform
- The project combines simulation, hardware implementation, and intelligent control exploration

Project Goal

- Reproduce the main workflow of DexHand as a course project
- Understand its mechanical, electronic, and control architecture
- Prepare for later integrated testing and intelligent control development

Main Work Packages

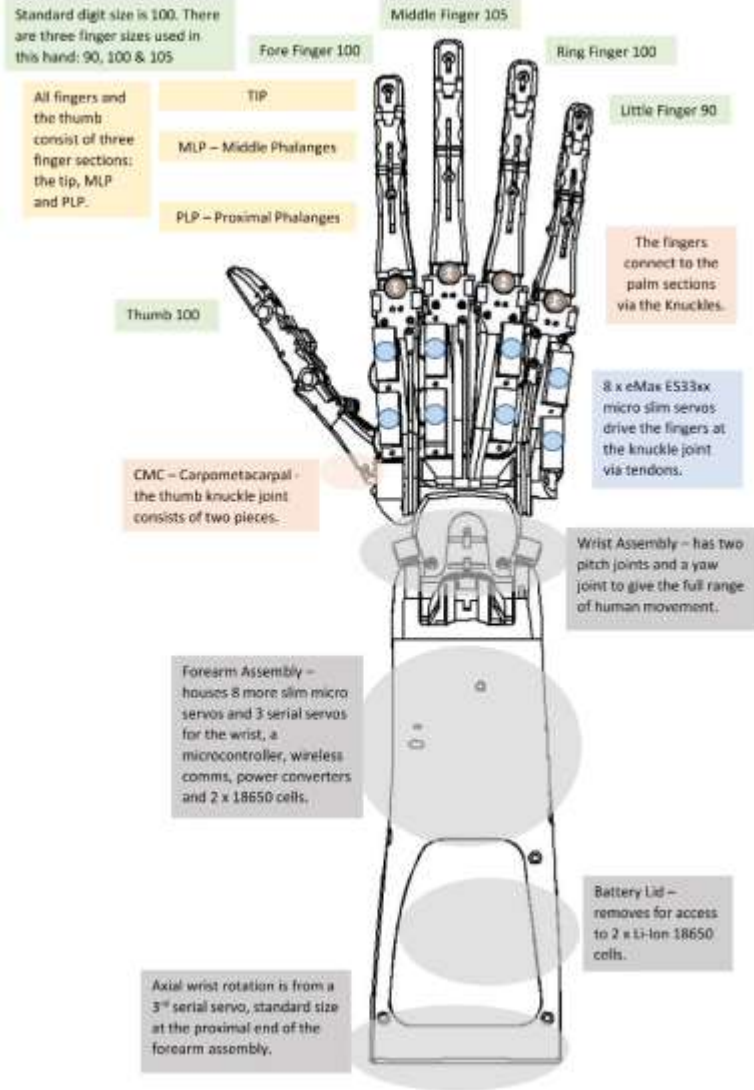
- Mechanical Assembly
- Electronics and Embedded Testing
- Simulation

Interim Progress Focus

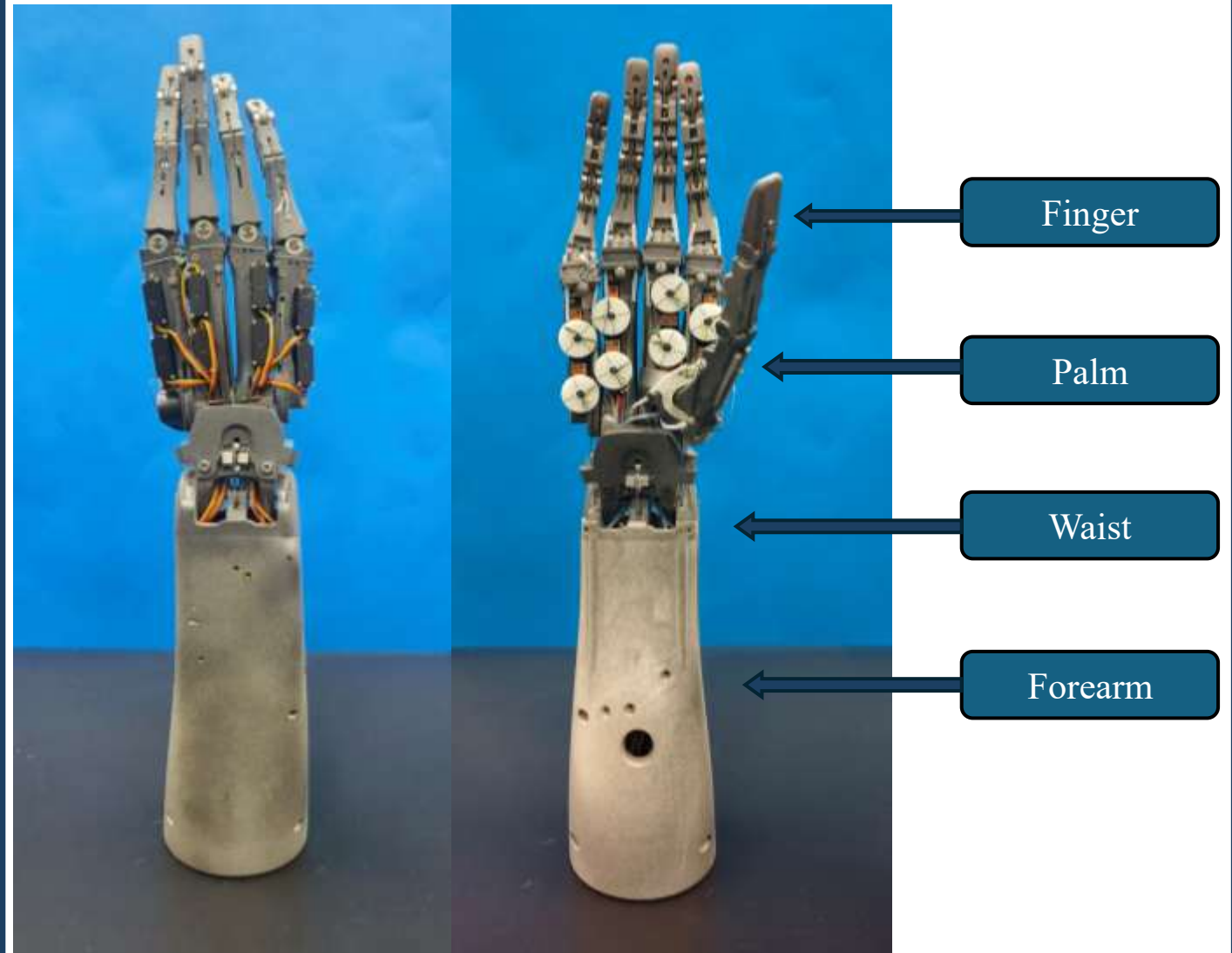
- ROS2 simulation workflow reproduced
- Hardware architecture and actuator selection analyzed
- Mechanical assembly and PCB preparation started



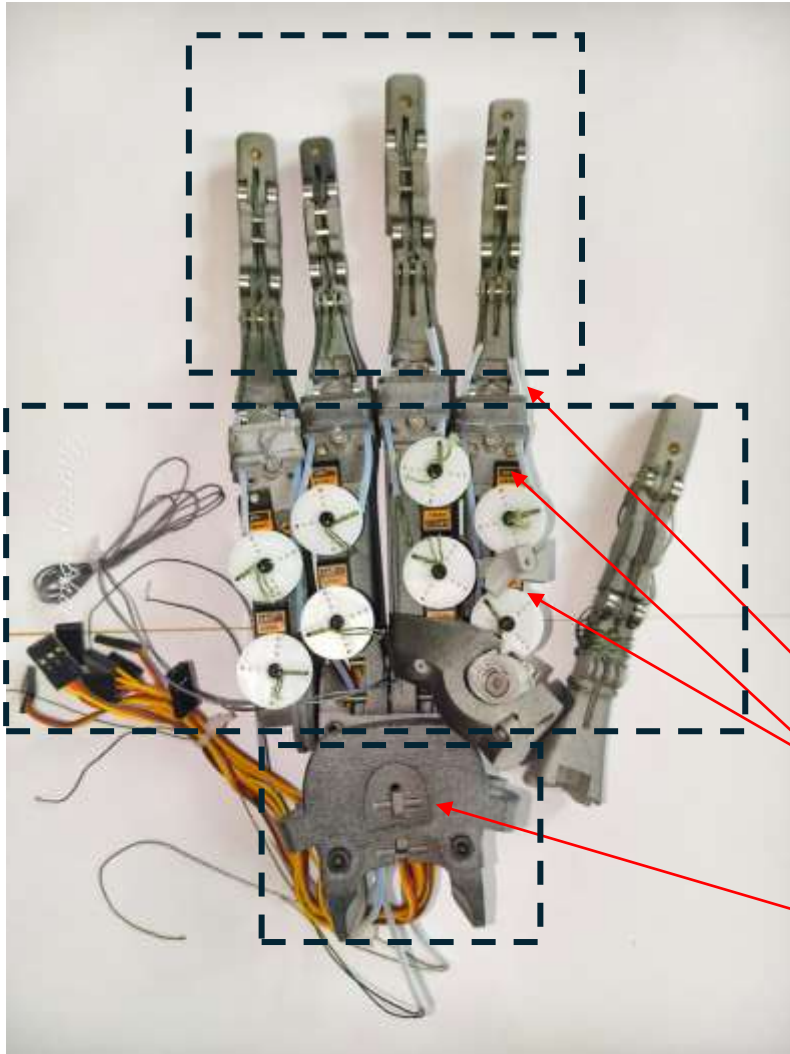
Ideal Effect



Mechanical structure



DOF- Finger & Palms & Wrist

**Finger**

Distal interphalangeal (DIP) joint flexion/extension → **1 DOF**

Proximal interphalangeal (PIP) joint flexion/extension → **1 DOF**

Metacarpophalangeal (MCP) joint flexion/extension and abduction/adduction → **2 DOF**

5 fingers × 4 DOF per finger = 20 DOF

**Palm**

The palm houses key components such as springs and servo motors.

Springs provide the restoring force for finger motion.

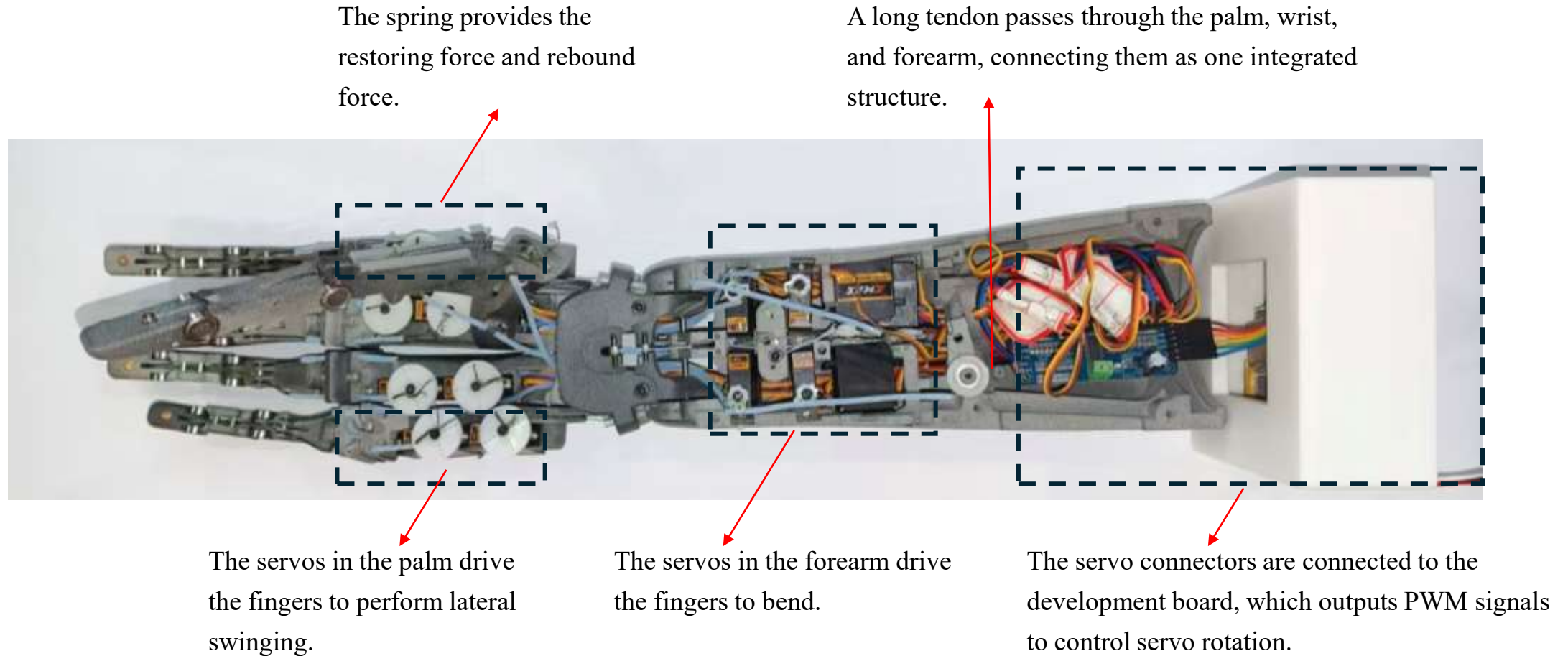
Tubes protect the tendons and prevent cable entanglement.

Each finger is actuated by two servo motors for lateral motion.

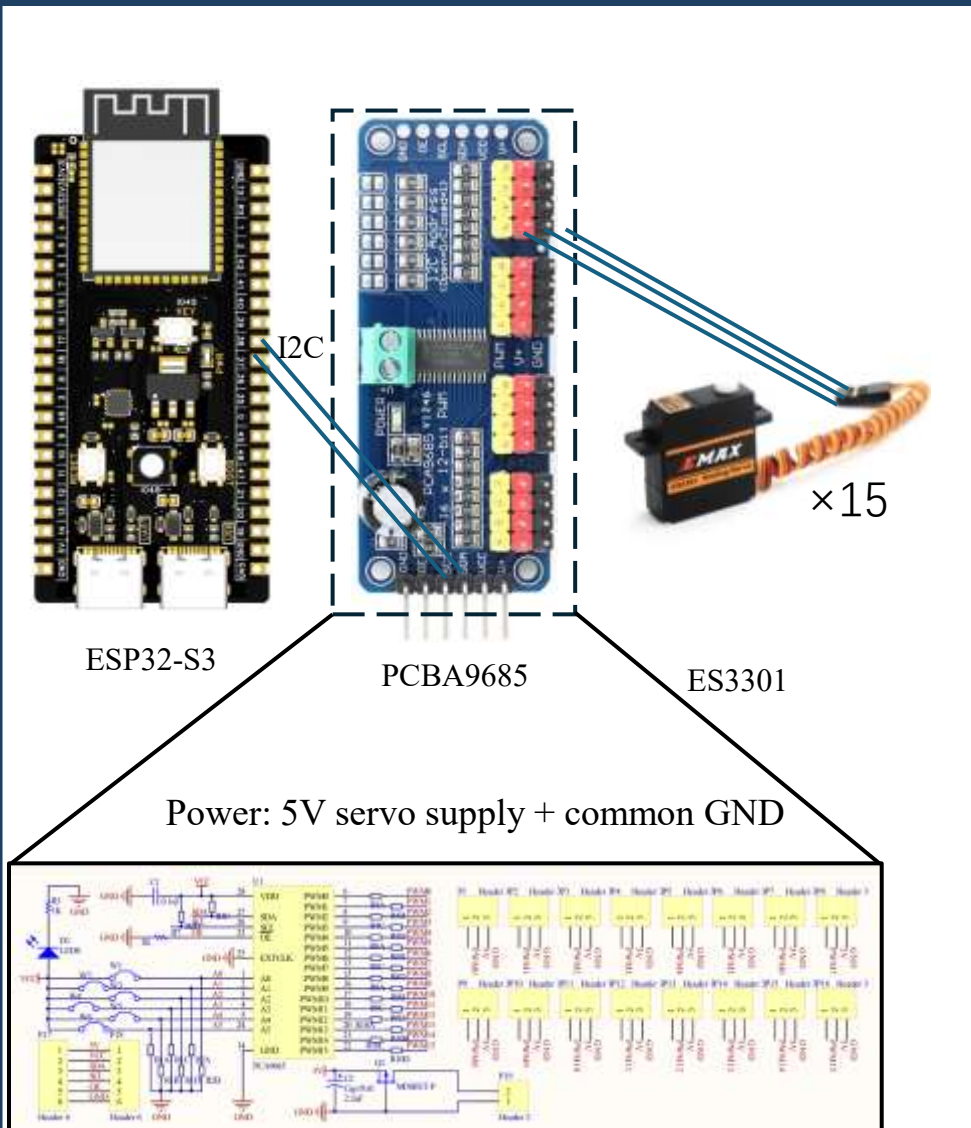
Wrist

The wrist supports **pitch** and **yaw**, corresponding to **1 + 1 = 2 DOF**.

Principle of tendon-driven actuation



ESP32-S3 Control Architecture



Firmware Embedded



Development Environment

- PlatformIO + VS Code
- Arduino framework on ESP32-S3
- WiFi networking and HTTP server
- PCA9685 driver library
- USB Serial debugging

Servo table mapping

Channel Mapping	
CH0 – CH2	Index finger: lower / upper / tip
CH3–CH5	Middle finger: lower / upper / tip
CH6–CH8	Ring finger: lower / upper / tip
CH9–CH11	Little finger: lower / upper / tip
CH12–CH14	Thumb: left / right / flexion

Initialize ESP32-S3

Connect To
WiFi

Start HTTP

Receive Servo
Command

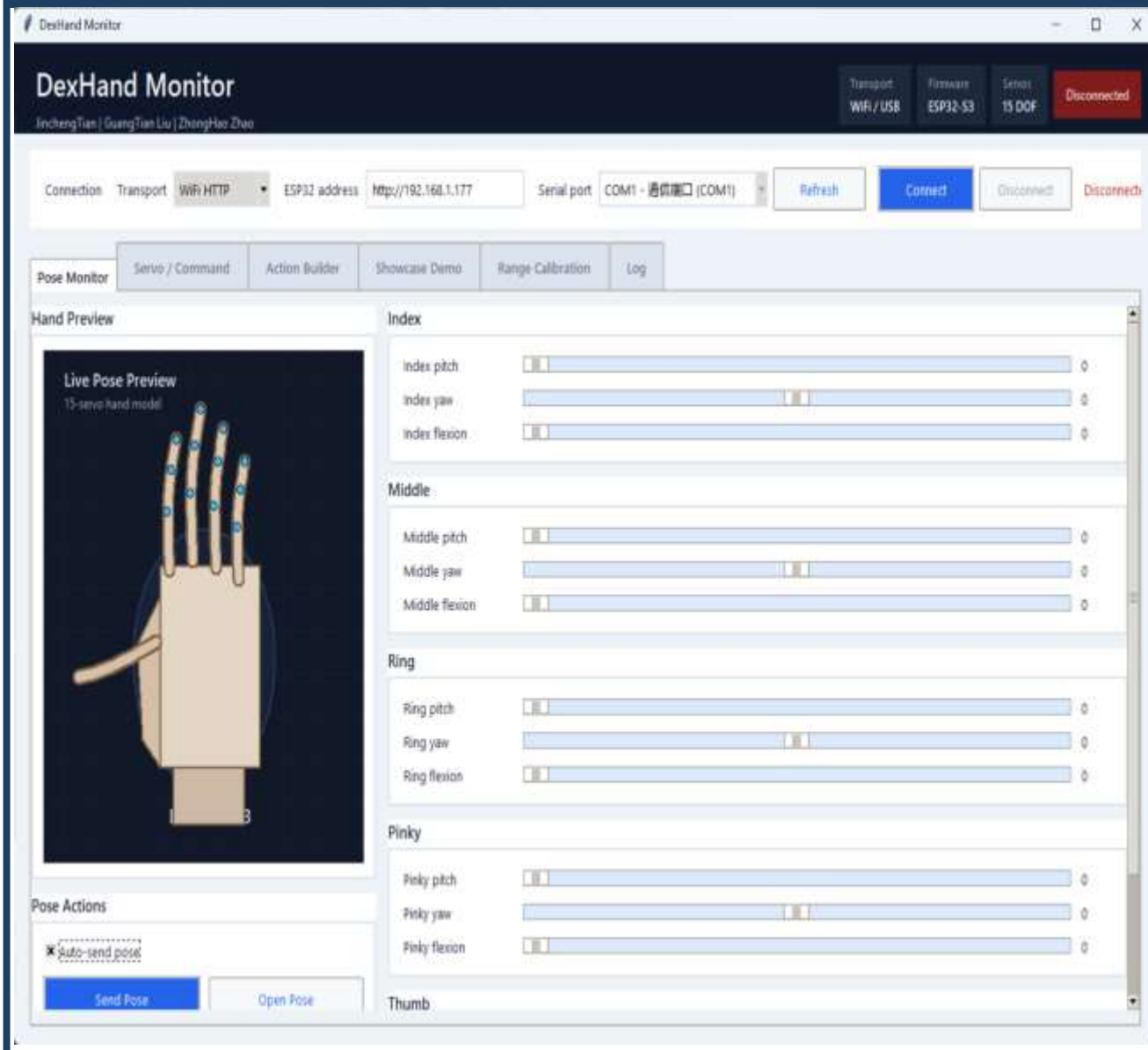
Check Servo Table

Limit Angle Range

Output PWM to
PCBA8965



Monitor Overview



Tkinter-Based Upper Computer

Tab Functions

- Pose Monitor** - Real-time hand pose control and live preview
- Servo/Command** - Independent single-servo control and command sending
- Action Builder** - Record current pose values for action save and replay
- Showcase Demo** - Demonstration playback and smooth transition control
- Range_Cal** - Calibrate initial and final positions for each servo
- Log** - Display communication logs and runtime feedback

Key Highlights

- Built with Tkinter for desktop-based monitoring and control
- Supports USB Serial and WiFi HTTP communication with ESP32-S3
- Demo 1:** Three Two One Fist sequential playback
- Demo 2:** Three > Yes smooth transition using linear interpolation

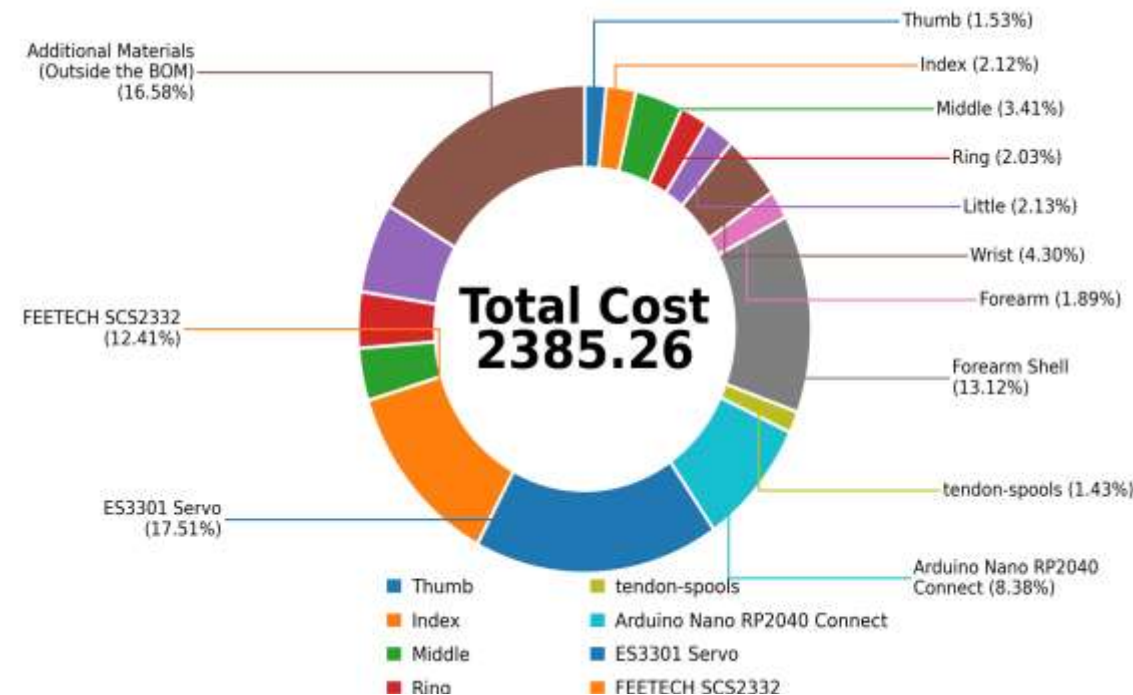


The monitor integrates real-time control, action recording, calibration, and demo playback in one desktop interface.



Components		Cost	Ratio	Satus
Finger	Thumb	36.41	1.53%	Purchased
	Index	50.68	2.12%	Purchased
	Middle	81.34	3.41%	Purchased
	Ring	48.41	2.03%	Purchased
	Little	50.77	2.13%	Purchased
Wrist		102.67	4.30%	Purchased
Forearm		45.05	1.89%	Purchased
Forearm Shell		313	13.12%	Purchased
tendon-spools		34	1.43%	Purchased
Arduino Nano RP2040 Connect		1*200=200	8.38%	Purchased
ES3301 Servo		16*26.1=417.6	17.51%	Purchased
FEETECH SCS2332		2*148=296	12.41%	Purchased
FEETECH SCS15		1*83=83	3.48%	Purchased
Fishing line No. 8		1*88.4=88.4	3.71%	Purchased
Other (See the BOM table for details)		142.38	5.97%	Purchased
Additional Materials (Outside the BOM)		About 395.55	16.58%	Purchased
Total		2385.26	100%	

Other: 4cm*6cm Arduino通用原型板/2.54 排母20pin/2.54 排针20pin/圆柱销套件/M2 螺丝及垫片/螺纹嵌件
 m2*4*3.5/12编100m灰色/6103滚珠轴承/263滚珠轴承/深沟球轴承6702ZZ/693zz 384/m3*60/m3*70





BOM&Materials

Prepared component list and purchased required mechanical, electronic, and wiring materials.

Mechanical Assembly

Completed DexHand finger, palm, wrist, and forearm assembly, tendon routing, and servo installation.

Electronics Integration

Built ESP32-S3 + PCA9685 based control platform and completed 15-servo wiring and power distribution.

Firmware Development

Developed ESP32-S3 firmware with Serial / WiFi communication, servo frame control, and safety limits.

Upper-Computer System

Developed a Tkinter-based monitor for pose control, servo command, action recording, calibration and log display.

Calibration & Demo

Calibrated servo ranges and show preset gesture playback with smooth transition and linear interpolation.



Demo



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